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A differential strategy for measurement of a static force in a single-point diamond cutting by a force-controlled fast tool servo

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Abstract

This paper presents a new strategy for applying a piezoelectric (PZT) force sensor to the measurement of static force in a force sensor-integrated fast tool servo (FS-FTS) system for single-point diamond cutting on planar brittle material substrates. A conventional PZT force sensor unit composed of a PZT force sensor and a charge amplifier cannot be employed for measurement of static force due to the discharging characteristics of PZT materials. In this paper, to realize the long-term stable measurement of static cutting force, the time constant of the PZT force sensor unit is set to be infinitely large by removing the feedback resistor from the charge amplifier. The influence of bias current leakage, which arises as a result of removing the feedback resistor, is compensated through obtaining the differential output of the two PZT force sensor units having identical charge amplifiers and identical thermal characteristics. After microgroove-cutting experiments to verify the stability of the PZT force sensor unit in the conventional FS-FTS unit, a theoretical model for the PZT force sensor unit is established. Some basic experiments are then carried out to verify the feasibility of the proposed differential measurement strategy. Furthermore, a modified FS-FTS unit with a pair of PZT force sensor units based on the proposed strategy is developed, and its effectiveness is demonstrated through experiments.

Keywords: fast tool servo, static force measurement, piezoelectric force sensor, differential measurement

(Some figures may appear in colour only in the online journal)

1. Introduction

In recent years, the demand for precision and accurate machining of components with freeform surfaces and/or microstructures has grown rapidly in many fields, such as in the optical industry and the semiconductor industry [1–4]. Since the performance of such components mainly depends on form accuracy and surface quality [5, 6], it is important to establish a machining process that can achieve both high accuracy and high throughput. Single-point diamond cutting with a fast tool

servo (FTS) is one of the candidate technologies for this purpose since the FTS can overcome the limited bandwidth of the tool feed motion in a machine tool [7, 8]. In addition, when machining brittle materials such as non-ferrous metals, crystals, polymers and ceramics, it is necessary to control the depth of cut precisely so that ductile cutting can be carried out [9–13]. However, it is sometimes difficult for systems based on conventional CNC machine tools to carry out ductile cutting on a non-planar surface since the cutting tool is controlled in position.

The above issue can be addressed by controlling the cutting tool in force [14–16]. Especially, in-process monitoring and feedback of the cutting force is an effective way to realize ductile cutting of brittle material substrates [17–19]. Some efforts have thus been made by using an atomic force microscope (AFM)-based system, where the AFM probe tip is employed as a cutting tool controlled in force [20–23]. Meanwhile, one of the drawbacks of the AFM-based machining is its low cutting force down to μN , which prevents the AFM-based machining from being applied to various machining applications. On the other hand, a piezoelectric (PZT) force sensor-based system can provide a high cutting force with high rigidity, while realizing highly sensitive force measurement [24–27]. The author's group has also developed a PZT force sensor-integrated fast tool servo (FS-FTS) system to deal with the difficulties of ultra-precision diamond cutting on a curved surface of a brittle material substrate [28, 29]. Furthermore, the FS-FTS system has also been employed to carry out not only force-controlled cutting but also on-machine profile measurement of microstructures [30]. However, the conventional FS-FTS system can only realize force-controlled cutting over a short period of several tens of seconds due to the inherent features of the PZT force sensor unit, which suffers from the discharging effect due to the finite insulation of PZT material. In addition, the PZT force sensor unit employed in the conventional FS-FTS system is sensitive to the change in temperature; these characteristics of the PZT force sensor unit prevent the FS-FTS system controlled in force from being applied to cutting operations over a large area. It is thus desirable to realize measurement of a static force over a long period by the PZT force sensor.

In this paper, the stability of the PZT force sensor unit is first verified through microgroove cutting experiments with the conventional FS-FTS system. After that, a theoretical model of the output signal from the PZT force sensor unit including the influences of the discharging effect, the bias current in the charge amplifier and the thermal stability of the overall PZT force sensor unit is established. Regarding the established theoretical model, a new strategy referred to as the differential force measurement strategy, in which a pair of PZT force sensors is employed to compensate for the aforementioned influences on the PZT force sensor unit, is proposed. A setup with the two PZT force sensor units is then developed to verify the feasibility of the proposed strategy. Furthermore, a new FS-FTS system with a pair of PZT force sensor units is developed, and its stability for the control of a diamond-cutting tool in force is verified through experiments.

2. Experimental verification of the stability of the conventional FS-FTS system

A diagram of the conventional FS-FTS system is shown in figure 1. The conventional FS-FTS unit is composed of a ring-shaped PZT actuator, a capacitive displacement sensor and a ring-shaped PZT force sensor. The motion axis of the PZT actuator and the measuring axis of the PZT force sensor are aligned to coincide with each other. The PZT force

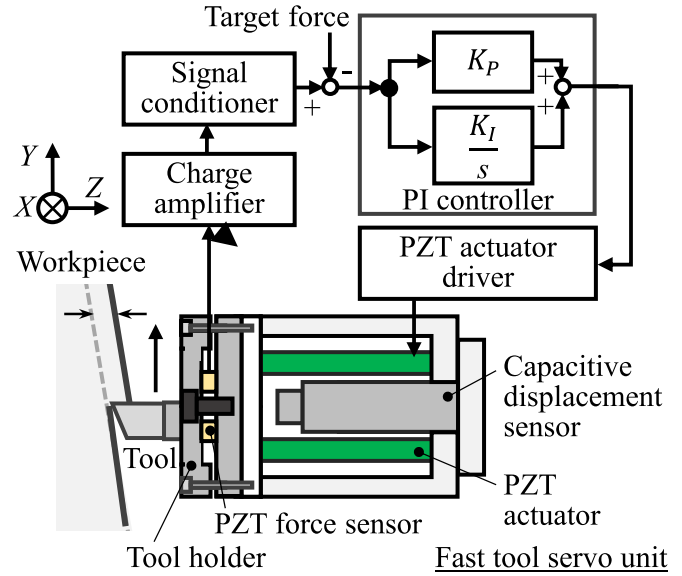


Figure 1. Diagram of the conventional force sensor-integrated fast tool servo (FS-FTS) unit and the control system.

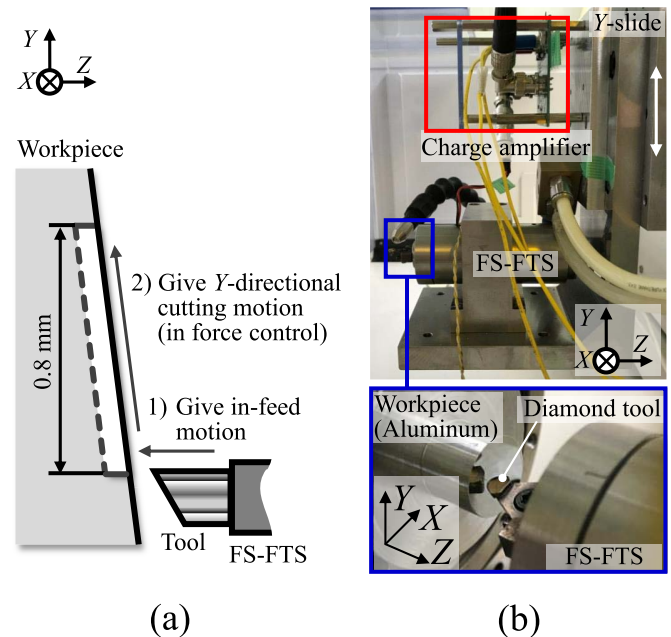


Figure 2. Setup of the microgroove cutting by using the FS-FTS. (a) Schematic of the microgroove cutting. (b) Photograph of the FS-FTS on the ultra-precision diamond turning machine.

sensor detects the thrust force (normal force) along the in-feed direction as the charge in the sensor. The current from the PZT force sensor, which is a consequence of the change in the charge of the PZT force sensor, can be converted into voltage output proportional to the integral of current. The voltage output from the charge amplifier is received by the signal conditioner and is then fed back to the PZT actuators in the FS-FTS unit through the PI controller and the PZT driver so that the thrust force can be kept constant as to the set reference force.

To evaluate the stability of the force feedback cutting by the conventional FS-FTS system, experiments were carried

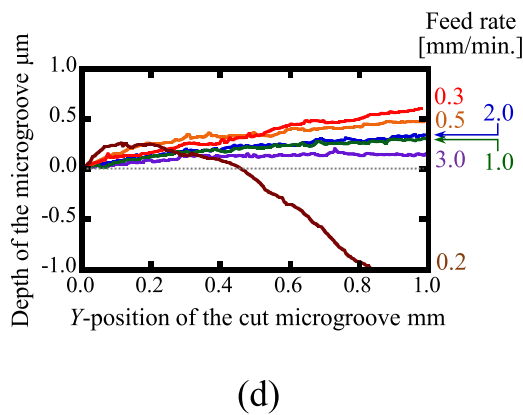
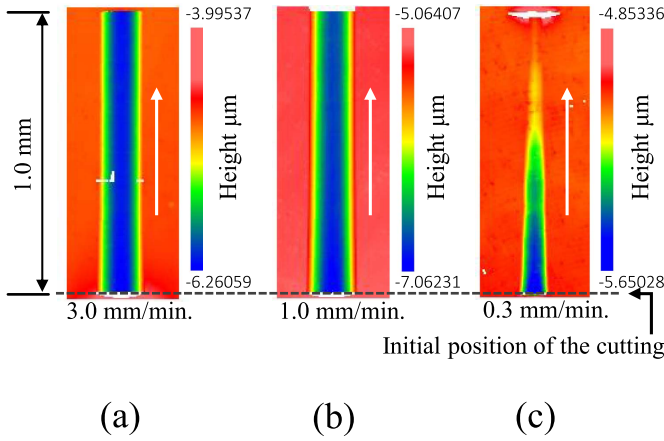


Figure 3. Profiles of the cut microgrooves measured by a three-dimensional optical profiler. (a) With a feed rate of 3.0 mm min^{-1} . (b) With a feed rate of 1.0 mm min^{-1} . (c) With a feed rate of 0.3 mm min^{-1} . (d) Depths of the cut microgrooves.

out. A schematic and a picture of the experimental setup are shown in figures 2(a) and (b), respectively. The FS-FTS unit was mounted on the Y-slide of a four-axis ultra-precision diamond turning machine (ULG-100, Toshiba Machine, Japan) in such a way that the motion axis of the FS-FTS unit became parallel with the Z-axis of the turning machine. An aluminum workpiece, whose surface was finished in advance of the experiments by using another diamond turning machine, was mounted onto the machine spindle with a vacuum chuck. A single-point diamond tool with a round nose radius of 2 mm and a clearance angle of 7° (ALMT C

Y-directional microgrooves were fabricated on the aluminum workpiece surface by controlling the Y-slide of the machine, while the workpiece was held stationary on the machine tool along the X- and Z-directions and the in-feed motion of the single-point diamond tool was controlled in force by the FS-FTS system. The reference force of the force feedback control in the FS-FTS system was set to be approximately 0.7 N. Six repetitive microgroove cuttings were carried out at different X-positions with different Y-directional feed rates ranging from 0.2 mm min^{-1} – 3.0 mm min^{-1} . Figures 3(a)–(c) show the profiles of the cut microgrooves with Y-directional feed rates of 3.0 mm min^{-1} , 1.0 mm min^{-1}

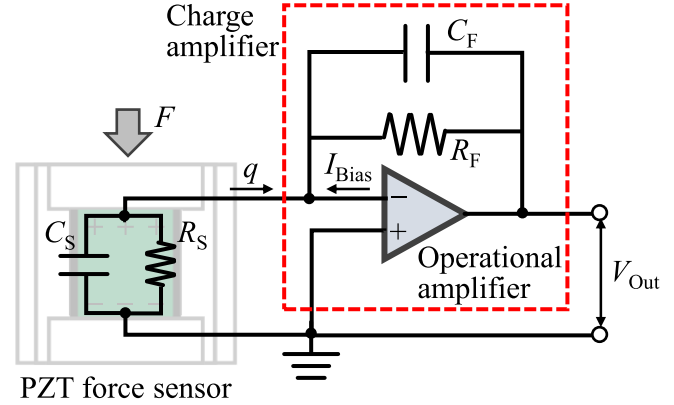


Figure 4. Circuit configuration of the PZT force sensor unit.

and 0.3 mm min^{-1} , respectively, measured by a commercial three-dimensional optical profiler (Zygo NewView7300). Figure 3(d) summarizes the change in the depth of each microgroove. As can be seen in the figure, the stability of the groove depth was found to worsen with the decrease of the feed rate; this was mainly due to the discharging effect of the PZT force sensor. This issue should be addressed to achieve cutting operation over a large area by force feedback control.

3. Establishment of the theoretical model of the PZT force sensor unit for the force feedback control

3.1. Theoretical model of the PZT force sensor unit

Aiming to improve the stability of the force feedback control in the FS-FTS unit, a theoretical model of the output signal from the PZT force sensor unit is established. Figure 4 shows the circuit configuration of the PZT force sensor unit composed of a PZT force sensor and a charge amplifier. The charge amplifier is composed of an operational amplifier with a feedback capacitor C_F and a feedback resistance R_F , and converts the sensor charge output Q_S into the voltage V_{Out} . When a force F is applied to the PZT force sensor along its measurement axis, the external charges on the sensor surface become greater than the internal charges, generating a total amount of Q_S of surface charges that can be expressed by the following equation [31]:

$$Q_S = F \cdot d_{33} \quad (1)$$

where d_{33} is the piezoelectric coefficient of the sensor material. Since the PZT sensor can be expressed by an equivalent circuit composed of a capacitor C_S and an insulation resistor R_S connected in parallel, the charge output q of the PZT force sensor can be expressed as a function of time t by the following equation [31]:

$$q = Q_S e^{-\frac{t}{R_S C_S}}. \quad (2)$$

The piezoelectric effect is usually adopted for dynamic cutting force measurement by utilizing the charge amplifier converting the high impedance electrical charge to a low impedance voltage output having large amplitude enough to

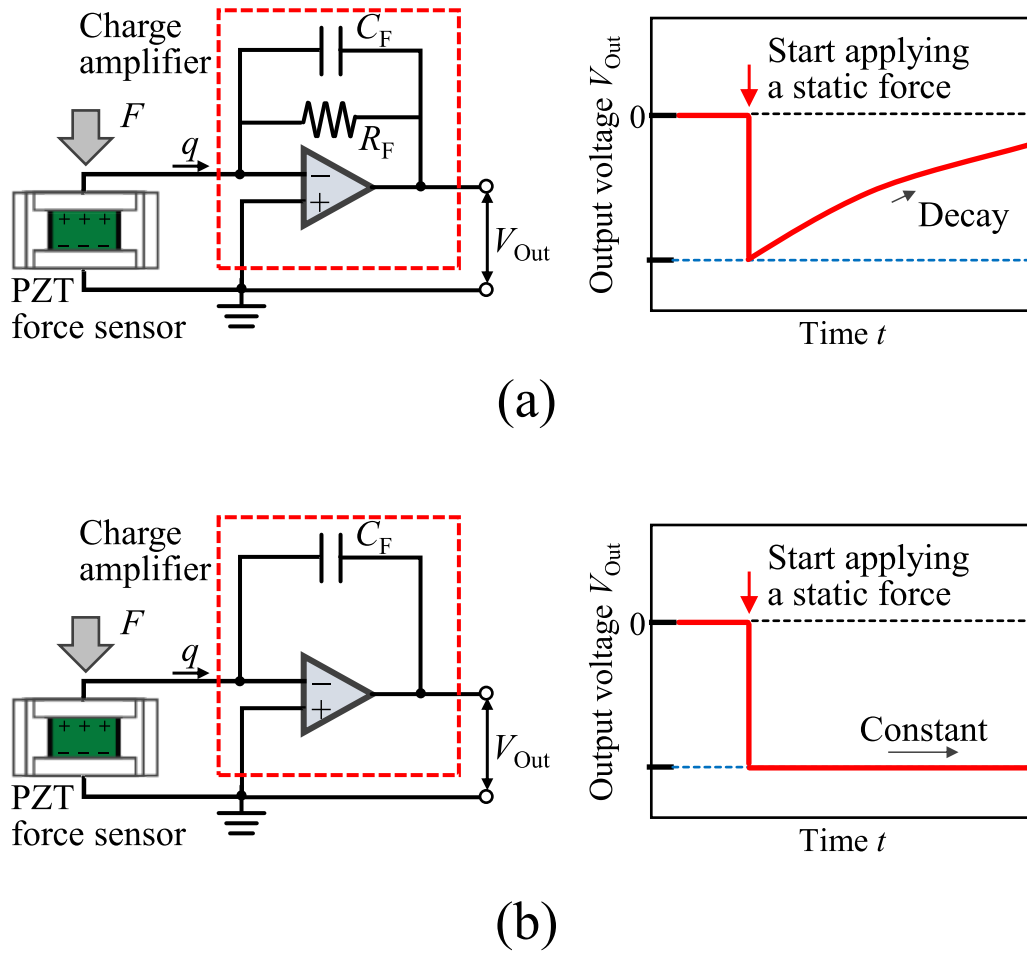


Figure 5. Circuit configurations of the PZT force sensor unit and corresponding voltage output. (a) With the feedback resistor. (b) Without the feedback resistor.

be measured. From the above equations, the voltage output of the charge amplifier V_{Out} can be expressed as follows:

$$V_{Out} = \frac{q}{C_F} e^{-\frac{t}{R_F C_F}} = \frac{Q_S e^{-\frac{t}{R_S C_S}}}{C_F} e^{-\frac{t}{R_F C_F}} = \frac{F \cdot d_{33}}{C_F} e^{-\left(\frac{t}{R_S C_S} + \frac{t}{R_F C_F}\right)}. \quad (3)$$

Since a general PZT force sensor has a large insulation resistor R_S up to hundreds of G Ω , equation (3) can be simplified as follows:

$$V_{Out} \approx \frac{F \cdot d_{33}}{C_F} e^{-\frac{t}{R_F C_F}} = \frac{F \cdot d_{33}}{C_F} e^{-\frac{t}{\tau_F}} \quad (4)$$

where τ_F , which is expressed as the product of R_F and C_F , represents the time constant of the charge amplifier. As can be seen in equation (4), the voltage output of the charge amplifier decays exponentially as a function of t even when a constant force F is applied to the piezoelectric force sensor as shown in figure 5(a); this phenomenon is referred to as the discharging effect [31]. This effect prevents the PZT force sensor unit from being employed for static cutting force measurement.

One of the possible solutions of employing PZT force sensor units to measure a static cutting force is to set the time constant τ_F to be large; by disconnecting the feedback resistor in the charge amplifier, the time constant becomes infinitely

large. This modification can decrease the influence of the discharging effect, as shown in figure 5(b). However, on the other hand, another problem arises; since there is no feedback resistor, the small bias current I_{Bias} from the operational amplifier in the charge amplifier continuously charges the feedback capacitor connected to the operational amplifier, resulting in instability of the voltage output of the PZT force sensor unit. Furthermore, the influences of the thermal instability of the PZT force sensor, as well as that of the charge amplifier, are issues to be addressed. In a practical case, the voltage output V of the PZT force sensor unit corresponding to the applied static force F can thus be expressed as follows:

$$V = \frac{F \cdot d_{33}}{C_F} + k \cdot I_{Bias} + f(T) \quad (5)$$

where k is a constant coefficient, and the term $f(T)$ represents the influence of the thermal drift of the overall system including the PZT force sensor and the charge amplifier with respect to the temperature T . As can be seen in equation (5), the terms related to the bias current I_{Bias} from the operational amplifier and the thermal instability of the system still remain in the charge amplifier output.

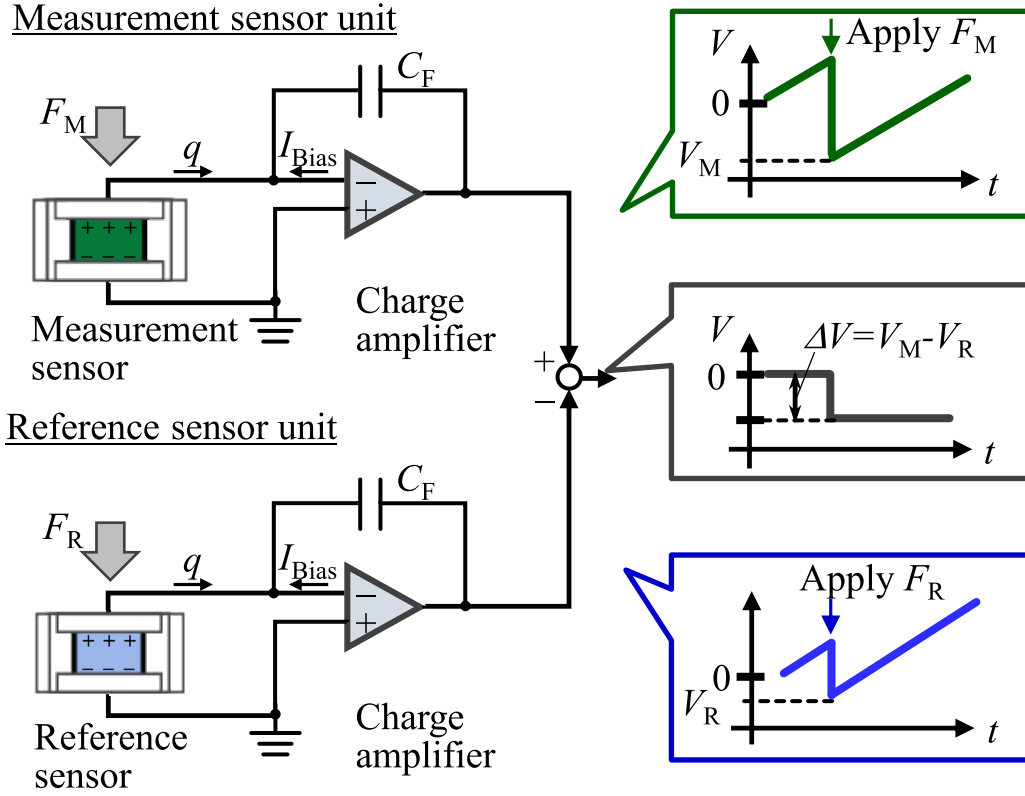


Figure 6. A schematic of the differential force measurement system with a pair of PZT force sensor units.

3.2. Differential force measurement strategy

To overcome the aforementioned problems, in this paper, a strategy employing a pair of PZT force sensors (referred to as the measurement PZT sensor and the reference PZT sensor in the following) is proposed. Figure 6 shows a schematic of the system for the proposed strategy, where one of the PZT sensor units is employed for the force measurement (measurement sensor unit), while the other is employed to compensate for the influences of the bias current from the amplifier and the thermal drift of the overall system (reference sensor unit). Now we consider the case where a static force F_M is applied to the measurement PZT sensor, while a static force F_R ($\neq F_M$) is applied to the reference PZT sensor. Assuming that the charge amplifier in the measurement sensor unit is identical to that in the reference sensor unit, and the specification of the measurement PZT sensor is identical to that of the reference PZT sensor, the voltage outputs from the measurement and the reference sensor units V_M and V_R , respectively, can be expressed as follows:

$$V_M = \frac{F_M \cdot d_{33}}{C_F} + K \cdot I_{Bias} + f(T) \quad (6)$$

$$V_R = \frac{F_R \cdot d_{33}}{C_F} + K \cdot I_{Bias} + f(T). \quad (7)$$

From these equations, the following equation can be obtained:

$$\Delta V = V_M - V_R = \frac{(F_M - F_R) \cdot d_{33}}{C_F}. \quad (8)$$

On the assumption that F_R is known as a given parameter, a static force F_M can be detected by taking the difference of the voltage outputs of the measurement and the reference sensor units.

4. Experiments for the verification of the proposed strategy with a pair of PZT force sensors

4.1. Investigation of the possible influencing factors on the stability of measurement

Experiments were carried out to demonstrate the effectiveness of the proposed strategy. In the following, the effect of extending the time constant by removing the feedback resistor in the charge amplifier, the influence of the bias current from the operational amplifier, and the influence of the thermal drift of the PZT force sensor units were investigated through experiments.

4.1.1. The effect of extending the time constant by removing the feedback resistor in the charge amplifier. First, the effect of extending the time constant of the charge amplifier was investigated. Figures 7(a) and (b) show a schematic and a photograph of the setup for the experiment. A PZT force sensor with an equivalent piezoelectric charge constant d_{33} of $600 \times 10^{-12} \text{ C N}^{-1}$ was fixed on a jig by a screw, and was connected to a charge amplifier composed of an operational amplifier (ADA4530-1, Analog Devices) with a low bias current of within $\pm 20 \text{ fA}$, a feedback capacitor C_F of

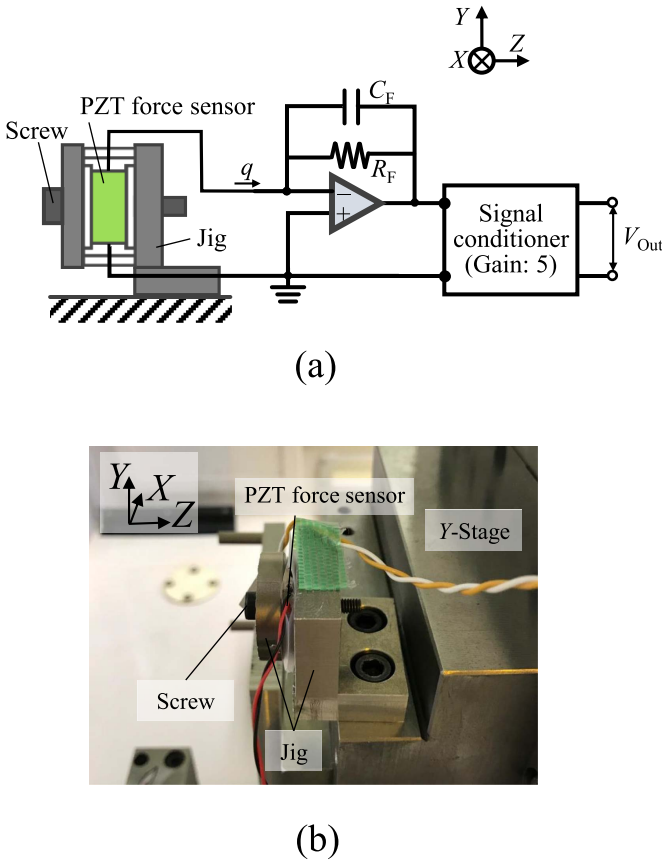


Figure 7. A setup for evaluating the time constant of the voltage output of a PZT force sensor unit. (a) A schematic of the setup. (b) A photograph of the PZT force sensor fixed on the jig.

47 nF and a feedback resistor. In the experiments, the voltage output of the PZT force sensor unit was monitored while the static force applied to the PZT force sensor was suddenly reduced by loosening the screw. Figure 8 shows the result. The voltage outputs with a feedback resistor of 1 G Ω and 10 G Ω as well as that without the feedback resistor (corresponding to $R_F = \infty \Omega$) are plotted in the figure. Time constants of the voltage output from the charge amplifier were evaluated to be 46.95 s and 417.58 s for the case with a feedback resistor of 1 G Ω and 10 G Ω , respectively. Each value was found to be close to the theoretical values (47 s and 470 s) obtained as the product of R_F and C_F . In addition, as predicted from the theory, a time constant for the case without the feedback resistor was confirmed to be infinitely large. Meanwhile, a slight increase in the voltage output of the PZT force sensor unit, which was due to the influences of the bias current from the operational amplifier and the thermal drift of the system, was observed.

4.1.2. The influence of the bias current from the operational amplifier. As the next step, the influence of the bias current from the operational amplifier was evaluated. Figure 9(a) shows a schematic of the setup for the experiment. Two charge amplifiers with identical characteristics were employed while the operational amplifiers in these charge amplifiers were driven by the same DC power supply (PMM18-2.5DU, Kikusui Electronics Corp., Japan). It should be noted that the PZT

force sensors were disconnected from the charge amplifiers so that only the influence of the bias current from the operational amplifier could be evaluated. Figure 9(b) shows the variation of the voltage output from each of the charge amplifiers, where the PZT force sensors were not connected. The differential output of these two voltage outputs obtained by an oscilloscope through a signal conditioner with a gain G of 5 is also plotted in the figure. A linear component and a cyclic nonlinear component can both be found in the voltage output of the charge amplifier. The linear component was due to the influence of the bias current I_{Bias} from the operational amplifier. Denoting the trend of the linear component in the voltage output from the charge amplifier as k , the bias current I_{Bias} can be calculated by the following equation:

$$I_{Bias} = \frac{C_F \cdot k}{G} \quad (9)$$

The trend k of the linear component in the voltage output from the charge amplifier for the measurement PZT sensor was evaluated to be $2.62 \times 10^{-7} \text{ V s}^{-1}$, corresponding to a bias current from the operational amplifier of 2.46 fA regarding a feedback capacitor C_F employed in the charge amplifier (47 nF). In the same manner, the bias current from the charge amplifier for the reference PZT sensor was evaluated to be 2.51 fA from an observed trend of $2.67 \times 10^{-7} \text{ V s}^{-1}$. The bias currents observed in the experiment were on the same level, and were within the specification of the operational amplifier (± 20 fA). The cyclic nonlinear component was mainly due to the instability of the DC power supply. Since both of the charge amplifiers were driven by the same DC power supply, the cyclic nonlinear components in the voltage outputs were synchronized with each other. Therefore, as can be seen in the figure, the influences of these nonlinear and linear components were found to be reduced by one order of magnitude in the differential voltage output.

4.1.3. The influence of the thermal drift of the PZT force sensor units. The thermal drift of the system based on the proposed strategy with a pair of PZT force sensors was also evaluated in experiments. Figure 10 shows a schematic of the setup where a pair of PZT force sensors with identical characteristics was employed aiming to compensate for the influences of the thermal drift. The two PZT force sensors were connected to identical charge amplifiers. During the experiment, the room temperature was intentionally changed with an air conditioner in the laboratory. Figure 11(a) shows voltage outputs of the measurement sensor unit and the reference sensor unit, as well as the change in the room temperature. A change in the voltage output in a period of 180 s was evaluated to be approximately 1.53 V, corresponding to a force F of 23.97 N. As can be seen from the result, the thermal stability of the PZT force sensors greatly influences static force measurement. Figure 11(b) shows the voltage outputs of the measurement sensor unit and the reference sensor unit with respect to the change in the temperature, which were re-plotted from the data shown in figure 11(a). Temperature sensitivity coefficients of the voltage outputs from the measurement

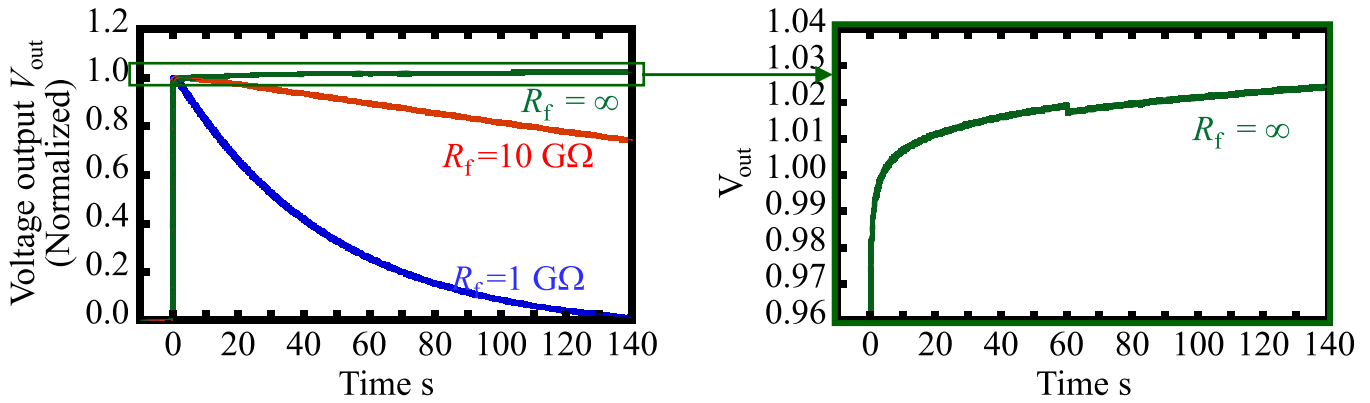


Figure 8. Variation of the voltage output of the PZT force sensor unit with different feedback resistors.

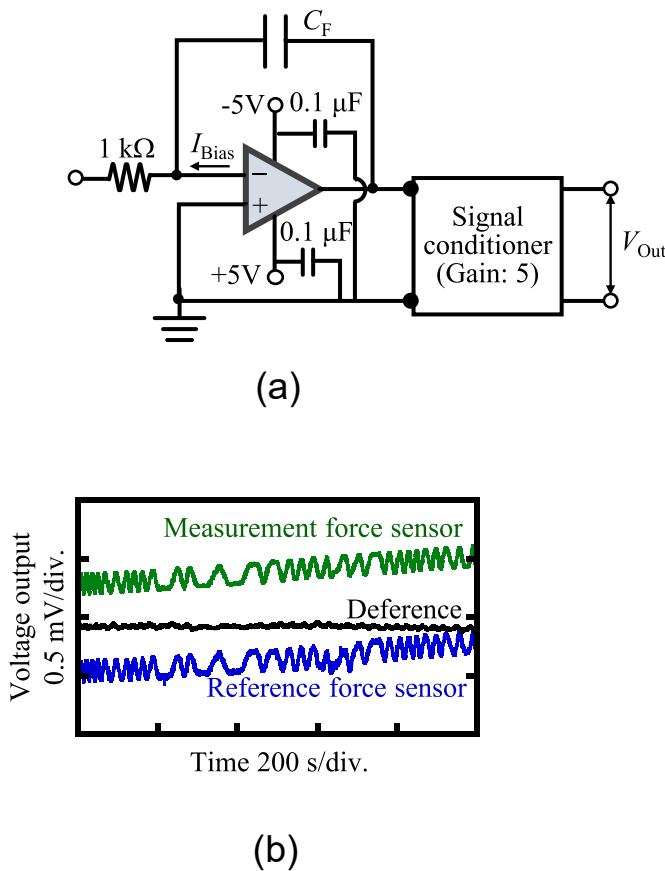


Figure 9. Influence of the bias current in the charge amplifiers in the measurement sensor unit and the reference sensor unit. (a) Circuit configuration of the force sensor unit for the experiment. (b) Voltage outputs from the sensor units.

sensor unit and the reference sensor unit after passing through the signal conditioner were evaluated to be $-3.314 \text{ V}^\circ\text{C}$ and $-3.246 \text{ V}^\circ\text{C}$, respectively, and were found to be close to each other. These identical temperature sensitivity coefficients of the PZT force sensor contribute to stabilizing the differential voltage output. A peak-to-peak deviation of the differential voltage output in a period of 180 s was evaluated to be 0.047 V, corresponding to the change in a force F of 0.736 N. By employing the proposed differential strategy,

the thermal stability of the force measurement (0.736 N over 180 s) was improved to be 3% of that in the conventional method where a sole PZT force sensor was employed (23.97 N over 180 s). Table 1 summarizes the thermal sensitivities of the voltage outputs of the two force sensors and that of the differential output. These experimental results demonstrated the feasibility of the proposed strategy in reducing the influences of the bias currents from the operational amplifier and the thermal drift of the overall system.

Following the above investigation, experiments were extended to the measurement of a static force based on the proposed differential strategy with the pair of PZT force sensors. In the experimental setup, the measurement axes of the measurement PZT sensor and the reference PZT sensor were aligned in the direction of gravity. By using weights, static forces F_M (1.45 N) and F_R (0.50 N) were applied simultaneously to the measurement PZT sensor and the reference PZT sensor, respectively. Figure 12 shows the voltage outputs of the two sensor units as well as the differential voltage output. As can be seen in the figure, a large drift component caused by the influences of bias currents from the operational amplifiers and the thermal drift of the system was observed in the voltage outputs of the PZT force sensor units. On the other hand, the differential output was almost kept constant over a period of 2000 s; this was much longer than the stable period of commercial piezoelectric force sensors on the order of several seconds. These results demonstrated the feasibility of the proposed differential strategy for the measurement of the static force.

4.2. Modification of the FS-FTS unit based on the proposed differential strategy with a pair of PZT force sensors

To further verify the feasibility of the proposed differential strategy, a modification is made to the conventional FS-FTS unit. Figure 13(a) shows the modified FS-FTS unit designed in this paper. Two ring-shaped PZT force sensors made of identical material but having different diameters are integrated into the modified FS-FTS unit. The PZT force sensors are placed between the tool holder and the PZT actuator, and the measurement axes of the force sensors are designed to be aligned to coincide with the axis of the FS-FTS unit so that the FS-FTS unit based on the proposed strategy becomes compact.

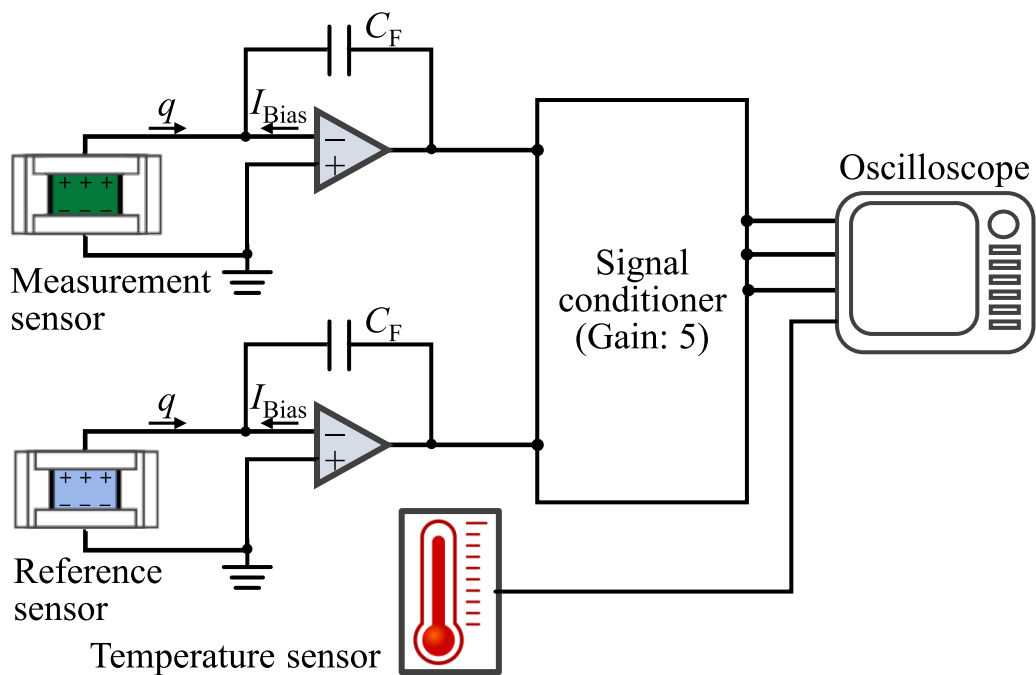
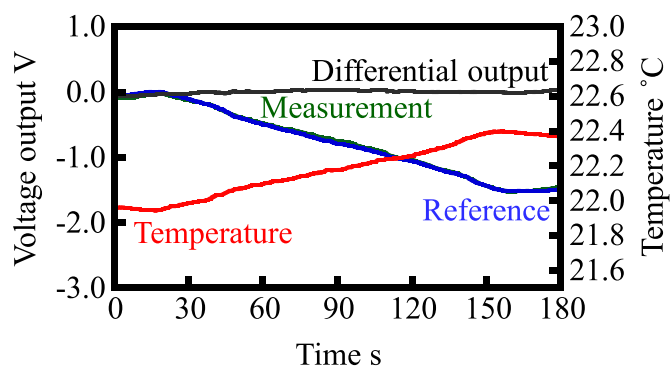
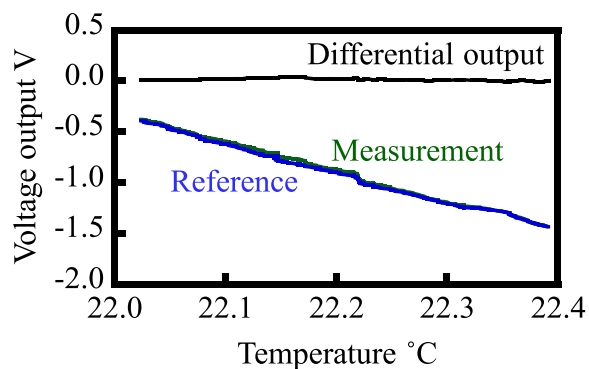


Figure 10. A schematic of the developed system based on the proposed differential strategy.



(a)

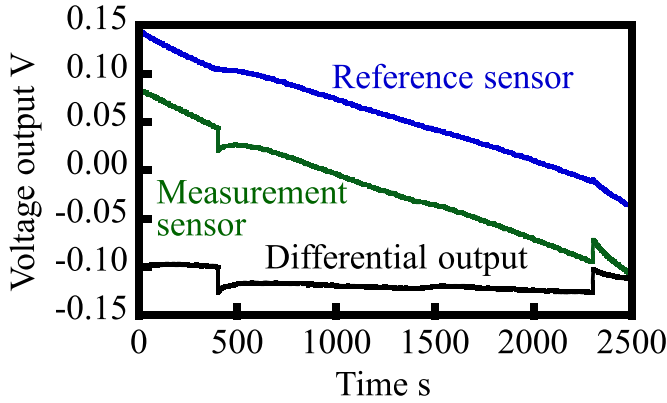


(b)

Figure 11. Thermal drift of the voltage outputs of the PZT force sensor units. (a) Variation of the outputs of the sensor units. (b) Relationship between the output and the temperature.

Table 1. Thermal stabilities of the force sensor outputs.

Measurement sensor output	Reference sensor output	Differential output
$-3.314 \text{ V/}^\circ\text{C}$ ($-51.917 \text{ N/}^\circ\text{C}$)	$-3.246 \text{ V/}^\circ\text{C}$ ($-50.852 \text{ N/}^\circ\text{C}$)	$-0.068 \text{ V/}^\circ\text{C}$ ($-1.065 \text{ N/}^\circ\text{C}$)

**Figure 12.** Off-machine static force measurement based on the proposed differential strategy.

This design is also expected to improve the thermal stability of the differential output, since the two force sensors are placed close to each other. In the setup, the thrust force F applied to the diamond tool on the modified FS-FTS unit is received by the two PZT force sensors. Denoting the cross-sectional areas of the measurement PZT sensor and the reference PZT sensor as A_M and A_R , respectively, the voltage output from each of the force sensor units can be expressed as follows based on equation (5):

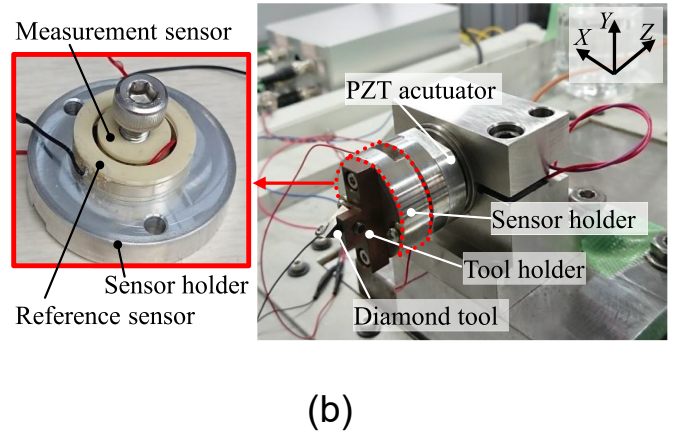
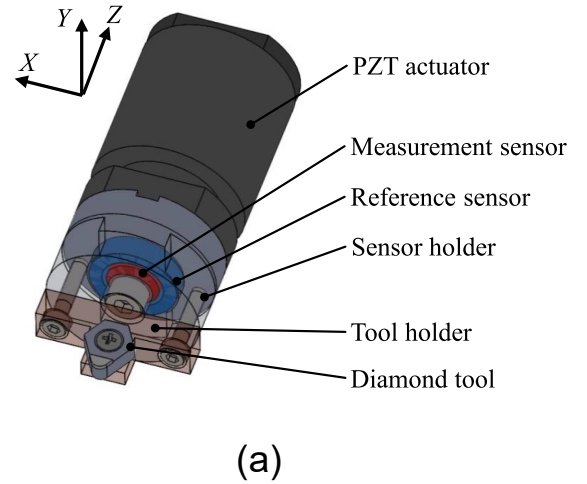
$$V_M = \frac{d_{33}}{C_F} \cdot \frac{A_M}{A_M + A_R} \cdot F + K \cdot I_{\text{Bias}} + f(T) \quad (10)$$

$$V_R = \frac{d_{33}}{C_F} \cdot \frac{A_R}{A_M + A_R} \cdot F + K \cdot I_{\text{Bias}} + f(T). \quad (11)$$

As can be seen in the above equations, the sensitivity of the PZT force sensor unit depends on the equivalent piezoelectric constant d_{33} identical to the material and the feedback capacitor C_F , regardless of the sensor size. Therefore, on the assumption that identical charge amplifiers are employed for both of force sensor units and the thermal drifts of the force sensor units are the same, the following equation can be obtained:

$$\Delta V = V_M - V_R = -\frac{d_{33}}{C_F} \frac{A_M - A_R}{A_M + A_R} \cdot F. \quad (12)$$

Since A_M and A_R are known as the design parameters of the FS-FTS unit, the static force F can be obtained by observing the voltage outputs V_M and V_R of the force sensor units, while reducing the influences of the bias currents in the charge amplifiers and the thermal drifts of the force sensor units.

**Figure 13.** The modified FS-FTS unit based on the proposed differential strategy. (a) Schematic of the modified FS-FTS unit. (b) Photograph of the modified FS-FTS unit with two PZT force sensors.

The feasibility of the modified FS-FTS was verified through experiments. Figure 13(b) shows a photograph of the modified FS-FTS with the ring-shaped PZT force sensors. Both force sensors were connected to different charge amplifiers having identical operational amplifiers whose feedback capacitor C_F and feedback resistor R_F were set to be 47 nF and $\infty \Omega$, respectively. The modified FS-FTS with a single-point diamond tool with a round nose radius of 2 mm and a clearance angle of 7° (ALMT Corp., Japan) was mounted on the four-axis ultra-precision diamond turning machine. An aluminum workpiece was mounted on the spindle of the machine. In the experiments, a static thrust force was applied to the diamond cutting tool by making the tool tip contact with a workpiece surface and held stationary along the in-feed direction. By using the PZT actuator in the modified FS-FTS unit, the tool tip was made to travel along the in-feed direction in a step of 1 μm . The contact between the diamond tool and the workpiece surface could be detected through monitoring the voltage outputs of the force sensor units. Once the contact was detected, both the aluminum workpiece and the diamond tool were held stationary while the variations of the voltage outputs

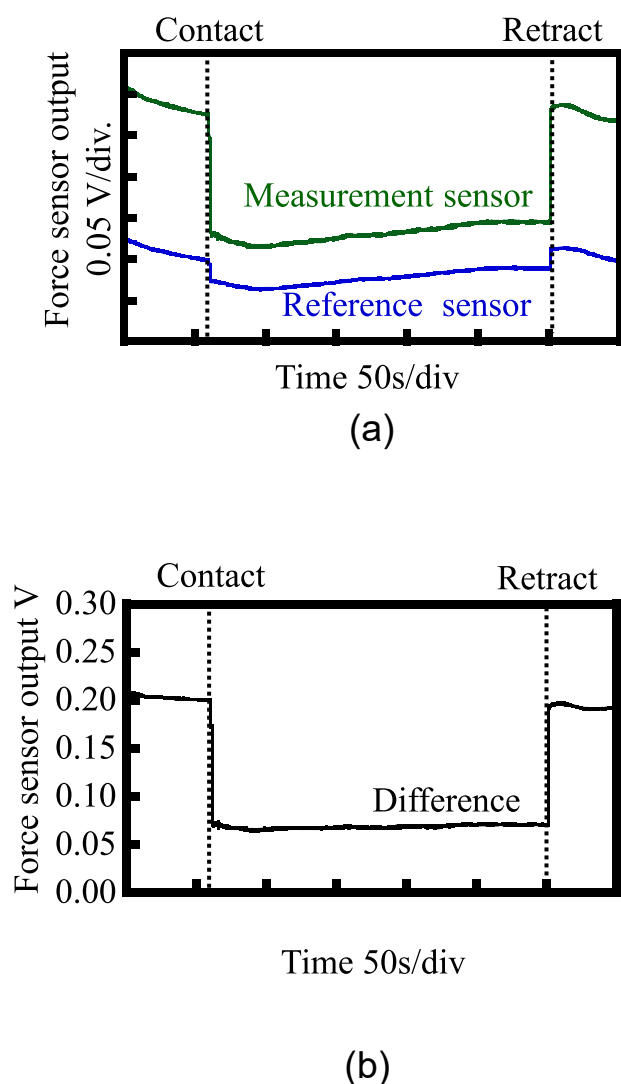


Figure 14. On-machine measurement of a static force applied to the modified FS-FTS unit based on the proposed differential strategy. (a) Voltage outputs of the sensor units during the contact between the tool tip and the workpiece surface. (b) The obtained differential output.

of the force sensor units were monitored. Figures 14(a) and (b) show the voltage outputs of the force sensor units and the differential voltage output, respectively, over a period of 240 s. The differential voltage output was found to be stable during tool–workpiece contact. The peak-to-peak deviation of the differential voltage output was evaluated to be 9.6 mV, corresponding to a deviation of the detected force F of 0.150 N. This result demonstrated the feasibility of measuring static force by using the PZT force sensors integrated into the modified FS-FTS unit.

5. Summary

A differential strategy has been proposed in this paper to achieve static force measurement in a force sensor-integrated fast tool servo (FS-FTS) system for controlling a cutting tool

in force. First, cutting experiments were done to investigate the stability of the PZT force sensor employed in the conventional FS-FTS system. Through the cutting process of several microgrooves with different feed rates ranging from 0.2 mm min^{-1} to 3.0 mm min^{-1} , the instability of the PZT force sensor in the conventional FS-FTS has been verified. To achieve the cutting operation over a large area by force feedback control, a new strategy referred to as the differential force measurement strategy employing a pair of PZT force sensors has been proposed. In the proposed strategy, the time constant of the voltage output of the force sensor unit, which is composed of a PZT force sensor and a charge amplifier, was set to be infinitely large by removing the feedback resistor from the charge amplifier. The influence of the bias current from the operational amplifier, which arises with the removal of the feedback resistor, was compensated through employing a pair of PZT force sensors and obtaining the differential output of the two sensors. This setup is also expected to reduce the influences of the thermal drifts of the PZT force sensor units, which is another issue to be addressed for the measurement of static force based on the PZT force sensor. A theoretical model for the PZT force sensor unit has been established, and the effectiveness of increasing the time constant of the voltage output of the PZT force sensor unit, the influences of the bias current in the charge amplifier as well as the thermal drift of the overall system have been investigated through some fundamental experiments. In addition, the result of the static force measurement based on the developed setup with identical PZT force sensors and charge amplifiers has demonstrated that the proposed differential strategy can realize static force measurement over a period of 2000 s with a deviation of the detected force of 0.736 N, which cannot have been realized by the piezoelectric force sensor in the conventional FS-FTS unit. Furthermore, a design modification has been made for the FS-FTS unit to integrate two PZT force sensors into the unit, and the experimental result has demonstrated that the modified FS-FTS unit based on the proposed differential strategy can carry out stable on-machine measurement of a static force over a period of 240 s.

It should be noted that this paper has focused on the proposal and demonstration of the newly proposed differential strategy. Further applications of the modified FS-FTS unit based on the strategy for the diamond cutting process controlled in force, as well as the uncertainty analysis of the force feedback control, will be carried out as future work.

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